

HDOC - 35

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Question

Write a menu-driven C program to read a variable-length square integer array and perform the following operations using separate functions:

1. Determine whether all the elements on its main diagonal are distinct.
2. Find the sum of its main diagonal elements.
3. Display its elements using a diagonal traversal. Begin with the element in the top-left corner and traverse successive diagonals alternately in the upward-right and downward-left directions.
4. Exit.

Program

```
#include <stdio.h>

void readMatrix(int n, int a[n][n]) {
    int i, j;
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
}

void checkDistinct(int n, int a[n][n]) {
    int i, j, distinct = 1;
    printf("Main diagonal elements: ");
    for (i = 0; i < n; i++)
        printf("%d ", a[i][i]);

    for (i = 0; i < n; i++)
        for (j = i + 1; j < n; j++)
            if (a[i][i] == a[j][j])
                distinct = 0;

    if (distinct)
        printf("\nAll main diagonal elements are distinct.\n");
    else
        printf("\nMain diagonal elements are not distinct.\n");
}

void diagonalSum(int n, int a[n][n]) {
    int i, sum = 0;
    for (i = 0; i < n; i++)
        sum += a[i][i];
    printf("Sum of main diagonal elements: %d\n", sum);
}

void diagonalTraversal(int n, int a[n][n]) {
    int d, i, j;
    printf("Diagonal traversal: ");

    for (d = 0; d <= 2 * n - 2; d++) {
        if (d % 2 == 0) {
            i = (d < n) ? d : n - 1;
            j = d - i;
            while (i >= 0 && j < n) {
                printf("%d ", a[i][j]);
                i--; j++;
            }
        } else {
            j = (d < n) ? d : n - 1;
            i = d - j;
            while (j >= 0 && i < n) {
                printf("%d ", a[i][j]);
                i++; j--;
            }
        }
    }
    printf("\n");
}

int main() {
    int n, choice;

    printf("Enter order of square matrix: ");
    scanf("%d", &n);
```

```

int a[n][n];
printf("Enter matrix elements:\n");
readMatrix(n, a);

do {
    printf("\n--- MENU ---\n");
    printf("1. Check distinct main diagonal elements\n");
    printf("2. Sum of main diagonal elements\n");
    printf("3. Diagonal traversal\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            checkDistinct(n, a);
            break;
        case 2:
            diagonalSum(n, a);
            break;
        case 3:
            diagonalTraversal(n, a);
            break;
        case 4:
            printf("Exiting program...\n");
            break;
        default:
            printf("Invalid choice\n");
    }
} while (choice != 4);

return 0;
}

```

Sample Output

Enter order of square matrix: 3

Enter matrix elements:

1 2 3

4 5 6

7 8 9

Enter your choice: 1

Main diagonal elements: 1 5 9

All main diagonal elements are distinct.

Enter your choice: 2

Sum of main diagonal elements: 15

Enter your choice: 3

Diagonal traversal: 1 2 4 7 5 3 6 8 9

Enter your choice: 4

Exiting program...

Result: The program successfully checks whether the main diagonal elements are distinct, calculates their sum, and displays the matrix using diagonal traversal.